

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

(192511088)

I GLEONA BLIS.J (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: gleona

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Q5 10 marks

Total Marks 38 /40

[Signature]
Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

QUESTION : 1

SUBNETTING FOR THREE LAB

PROBLEM UNDERSTANDING

Given network : $192.168.20.0/24$

Three labs require different subnet sizes.

- lab A $\rightarrow /24$

- lab B $\rightarrow /27$

- lab C $\rightarrow /28$

We need to find ;

- 1) Network address.
- 2) Host IP range
- 3) Broadcast address.
- 4) Usable hosts

Method Selection

Use the formula :

$$\text{Usable hosts} = 2^{\text{host bits}} - 2$$

- $/26 \rightarrow 32 - 2 = 30$ host bits

- $/27 \rightarrow 8 - 2 = 6$ host bits

- $/28 \rightarrow 4 - 2 = 2$ host bits

CALCULATIONS:

$$\text{LAB A: } 2^6 - 2 = 62 \text{ hosts}$$

$$\text{LAB B: } 2^5 - 2 = 30 \text{ hosts}$$

$$\text{LAB C: } 2^4 - 2 = 14 \text{ hosts}$$

Interpretation:

The /26, /27 and /28 subnets provide 62, 30, and 14 usable IP addresses, respectively. Subnetting allows the university to divide one /24 network efficiently among different labs.

Final Answer: Lab A $\rightarrow$.0/26, Lab B $\rightarrow$.64/27
Lab C $\rightarrow$.96/28.

QUESTION : 2 [DETERMINE SUBNET OF TWO SYSTEMS]

PROBLEM UNDERSTANDING:

SYSTEM 1:

- IP = 192.168.1.130
- Mask = 255.255.255.192

SYSTEM 2:

- IP = 192.168.3.190
- Mask = 255.255.255.192
- CIDR = /26

Method selection:

For /26 :

Block size :

$$256 - 192 = 64$$

Therefore, subnet ranges are :

- 0 - 63
- 64 - 127
- 128 - 191
- 192 - 255

Solution process

System 1 - 192.168.1.130

130 belongs to the range 128 - 191.

THEREFORE

- Network address = 192.168.1.128
- Host range = 192.168.1.129 - 192.168.1.190
- Broadcast = 192.168.1.191

CALCULATION:

Host bits :

$$32 - 26 = 6$$

Usable host :

$$2^6 - 2 = 62$$

INTERPRETATION:

Both systems belong to the same /26 subnet, so they can communicate directly within the same subnet without requiring a router.

QUESTION - 3

DEPARTMENT SUBNET ALLOCATION:

PROBLEM UNDERSTANDING:

Given network:

172.16.0.0 /16

Departments:

- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

1) HR - /25

subnet mask = 255.255.255.128

Block size = 128

- Network = 172.16.0.0
- Host range = 172.16.0.1 - 172.16.0.126
- Broadcast = 172.16.0.127

sales - /26

Next available address = 172.16.0.128

subnet mask = 255.255.255.192

- Network = 172.16.0.128
- Host range = 172.16.0.129 - 172.16.0.190
- Broadcast = 172.16.0.191

Usable host :

$$2^6 - 2 = 62$$

3) IT - /27

Next available address = 172.16.0.192

subnet mask = 255.255.255.224

- Network = 172.16.0.192
- Host range = 172.16.0.193 - 172.16.0.223

Usable hosts : $2^5 - 2 = 30$

INTERPRETATION :

The organisation has successfully allocated different subnet sizes according to the departmental requirements. This provides efficient IP address utilization while leaving a large portion of the /16 block available for future departments.

QUESTION: 4 /27 SUBNETTING:

PROBLEM UNDERSTANDING:

Given :

172 . 16 . 10 . 0/24

subnet mask:

/27 = 255 . 255 . 255 . 224

We need to find :

- Number of students
- Usable host per subnet
- Subnet containing 172 . 16 . 10 . 78

METHOD SELECTION:

Borrowed bits :

$$27 - 24 = 3 \text{ bits}$$

Number of students

$$2^3 = 8 \text{ subnets}$$

Host bits :

$$32 - 27 = 5 \text{ bits}$$

Usable hosts :

$$2^5 - 2 = 30 \text{ host / subnet}$$

Subnet Ranges :

Block size :

$$256 - 224 = 32$$

FINAL ANSWERS :

- Number of subnets = 8
- Usable hosts / subnets = 30.
- 172.16.10.78 belongs to 172.16.10.64/27
- Network = 172.16.10.64
- Host range = 172.16.10.65 - 172.16.10.94
- Broadcast = 172.16.10.95

• .95 is in the same subnet range but is the broadcast address, not a usable host address

INTERPRETATION:

The /27 mask divides the network into 8 equal subnets, each supporting 30 usable devices. The IP .78 is a valid host, whereas .95 is reserved as the broadcast address.

QUESTION: 5 IPV6 ADDRESSING:

Given:

2001 : 0db8 : 0000 : 0000 : 0000 : ff00 : 0042 :
8329

with /64 prefix.

step 1 compressed address:

2001 : db8 :: 42 : 8329

step 2 - Expanded Address:

2001 : 0db8 : 0000 : 0000 : 0000 : ff00 : 0042 :
8329

step 3 - Network Prefix:

2001 : db8 :: /64

step 4: Host Addresses

Host bits = $128 - 64 = 64$ bits

Possible addresses : 2^{64}

18, 446, 744, 073, 709, 551, 616

INTERPRETATION:

- The /64 IPv6 network provides a very large number of possible addresses.